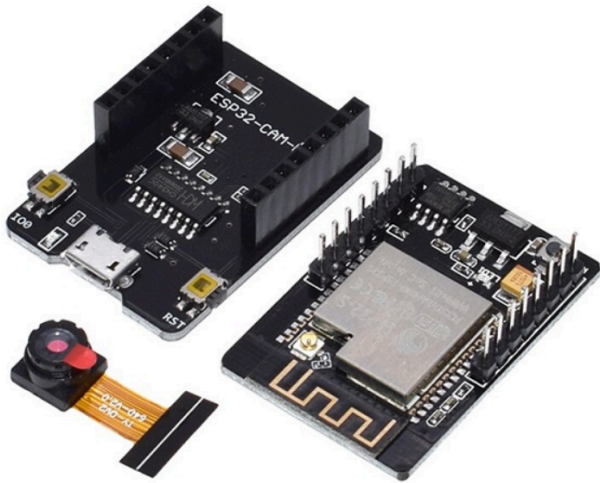


ESP32-CAM



Operating Voltage: 5V (Input), 3.3V Logic

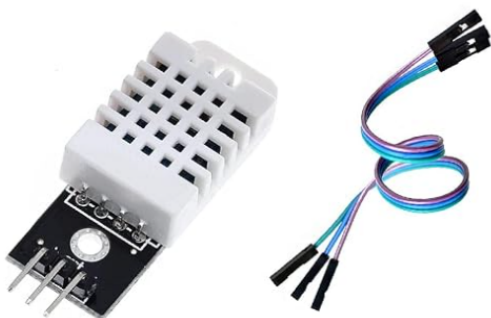
Function

The **ESP32-CAM** is the brain of the entire system. It collects data from all sensors, captures images of the plant, processes the data, controls the water pump and LED grow light, and sends information to the IoT dashboard through Wi-Fi.

Responsibilities

- Captures plant images
- Reads all sensor values
- Connects to Wi-Fi
- Uploads sensor data
- Controls the water pump
- Controls the grow light
- Runs AI/decision logic
- Sends notifications

DHT22 Temperature & Humidity Sensor



Operating Voltage: 3.3V–5V

Function

Measures the surrounding environmental conditions.

Measures

- Air Temperature (°C)
- Relative Humidity (%)

Why it is important

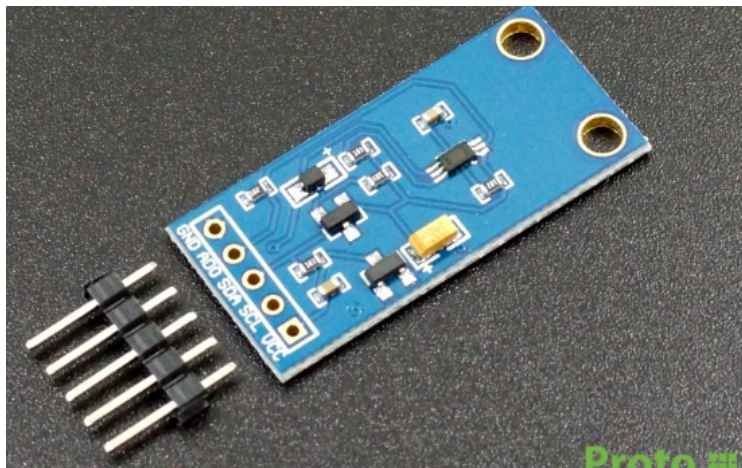
Plants require specific temperature and humidity ranges for healthy growth.

Example:

- High temperature → Plant stress
- Low humidity → Faster water loss
- High humidity → Increased disease risk

The ESP32-CAM uses this information to determine whether environmental conditions are suitable for plant growth.

BH1750 Digital Light Sensor



It measures how much visible light falls onto a specific surface area. One lux equals one lumen of light spread over a one-square-meter area.

Operating Voltage: 3.3V

Function

Measures the intensity of natural light in **Lux**.

Why it is important

Plants require sufficient sunlight for photosynthesis.

Example:

Morning

8000 Lux

No need for grow light.

Cloudy weather

2000 Lux

Turn ON grow light.

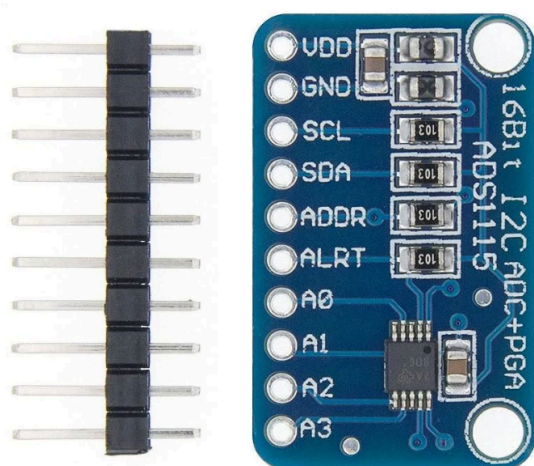
Night

0 Lux

Grow light automatically ON.

The ESP32-CAM compares the measured light intensity with a preset threshold and controls the grow light accordingly.

ADS1115 (16-bit Analog-to-Digital Converter)



Function

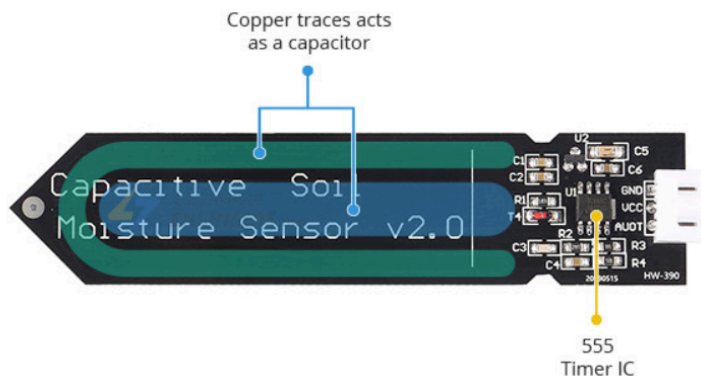
The ESP32-CAM has limited analog input capability. The ADS1115 converts analog sensor signals into high-resolution digital data that the ESP32-CAM can process.

Why it is used

- High accuracy
- Stable readings
- Supports multiple analog sensors
- Improves measurement precision

It acts as a bridge between analog sensors and the ESP32-CAM.

Capacitive Soil Moisture Sensor



Operating Voltage: 3.3V

Function

Measures the moisture content present in the soil.

Unlike resistive sensors, the capacitive sensor is corrosion-resistant and provides more stable readings over time.

Working

Wet Soil

High moisture value

Pump remains OFF

Dry Soil

Low moisture value

Pump turns ON automatically.

This helps prevent both underwatering and overwatering.

White Full-Spectrum LED Grow Light

Operating Voltage: 5V

Function

Provides artificial light when natural sunlight is insufficient.

The white full-spectrum LED emits wavelengths suitable for photosynthesis and healthy plant growth.

When it operates

- Night
- Cloudy weather
- Indoor environments
- Low light conditions

The ESP32-CAM automatically switches it ON or OFF based on the BH1750 sensor readings.

Mini DC Water Pump



Operating Voltage: 5V

Function

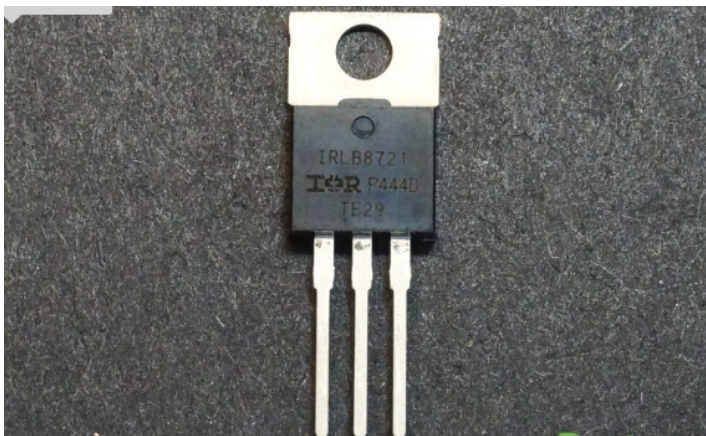
Supplies water automatically to the plant.

The pump activates only when the soil moisture level falls below the preset threshold.

Advantages

- Saves water
- Prevents plant drying
- Reduces manual watering
- Enables automated irrigation

MOSFET Driver Module



perating Voltage: 5V Logic

Function

The ESP32-CAM cannot directly power high-current devices like pumps or grow lights. The MOSFET acts as an electronic switch.

Controls

- Mini Water Pump
- White LED Grow Light

Advantages

- High efficiency
- Low heat
- Fast switching
- Low heat generation
- Supports PWM for brightness control
- Longer component life

Without a MOSFET, connecting these devices directly to the ESP32-CAM could damage the microcontroller.